Guidance, Not Adaptation: What Makes an Open Text-to-Music Model Sing isiZulu

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Abstract. Open text-to-music models claim support for fifty or more languages. To our knowledge, none has been evaluated on an African language. We adapt ACE-Step 1.5, a 2B-parameter flow-matching diffusion transformer, to isiZulu using 10.95 hours across ten genre families: chiefly Maskandi, Kwaito, Amapiano, Isicathamiya, Mbaqanga and Gqom. The corpus is itself largely model-generated: an artist drafted the isiZulu lyrics with a language model, corrected them, then produced the audio by prompting a music generation system. This is adaptation on synthetic in-language data, a regime increasingly common where field recordings are scarce.

Three findings survive that provenance, and none of the failures behind them raised an error. First, guidance rather than the adaptation controls whether the model sings the requested isiZulu words. Composite lyric error differs by -0.437 between guidance 1 and 7 ($d_z = -2.57$, sign-flip permutation $p < 10^{-5}$, $n = 60$ prompts), with only 6% of it mediated by vocal prominence. Second, the default distilled checkpoint overrides the guidance argument to 1.0. This documented behaviour is logged but not surfaced to the caller, so sweeping guidance on a default install sweeps an inert parameter. Third, the caption-retrieval control we built to validate the genre metric was itself broken in three independent ways. Each pushed towards a false negative and none raised an error. Corrected, the correct caption ranks 6.6 places above a mismatched one (95% CI $[3.0, 10.1]$, $p = 0.0008$, clustered over 63 captions), where the binarised statistic we first used found nothing.

Keywords: Text-to-music generation · isiZulu · low-resource languages · evaluation validity · classifier-free guidance

1 Introduction

ACE-Step 1.5 [1] advertises "strict adherence to prompts across 50+ languages". Its technical report names four (Chinese, English, Japanese and Thai), evaluates on two, and never enumerates the rest. The claim therefore cannot be falsified as stated. We can find no published evaluation of any African language on this or any comparable open model.

Eight comparable open systems make the same claim [2–9]: a language count in the abstract, but no per-language evaluation. All score style with an audio–text embedding, most often LAION-CLAP [11], trained predominantly on Western commercial music and English captions. We know of no published negative control establishing that such a score discriminates on non-Western material.

African NLP has an established answer: audit the resource before trusting it. Kreutzer et al. [14] found widely used multilingual corpora with no usable text in the languages they claimed. Later work shows automatic metrics ranking systems opposite to human judgement for a low-resource language while agreeing for a higher-resource one [15]. A confound as simple as document length can also reverse a conclusion [19]. We apply that discipline one level down, to the instrument rather than the corpus.

Neither gap has been closed, so we address both at once. isiZulu is a clean probe: the model's cross-lingual mechanism is stochastic Romanisation of non-Roman scripts, inert for a Latin-script language. Yet isiZulu sits far from the evaluated set phonologically and in its conjunctive orthography. We adapt ACE-Step 1.5 to it on 10.95 hours the artist generated from her own lyrics. The reference text is therefore the exact prompt, not a transcription, and lyric accuracy needs no annotator. We show three things.

Guidance, not the adaptation, controls whether the model sings the requested words, an effect the architecture predicts (§3.2). The default distilled checkpoint discards the guidance argument, so anyone benchmarking a default install sweeps an inert parameter (§3.2). Finally, the control we built to test both was itself broken three ways. Each failure pushed towards a false negative; repairing it turns an apparent instrument failure into a usable measurement (§3.3).

2 Methodology

2.1 Corpus

The corpus comprises 175 clips from 71 songs, covering 10.95 hours across ten genre families. A single isiZulu artist wrote the lyrics with a language model's help and corrected them for grammar and idiom. She then produced the audio by prompting a music generation system with those lyrics and a style caption: *model-generated audio carrying artist-authored isiZulu lyrics*, not field recordings. The two batches differ in caption convention (Table 1).

We group songs using both filename stem and a normalised lyrics hash. We apply this rule under transitive closure because 38 duplicate groups contain one song under unrelated filenames. Genre labels were assigned per song by ear; multi-genre captions are accurate because these genres genuinely blend in South African practice. Splits are song-level and genre-stratified, with 56 songs (132 clips) for training and 15 (43 clips) held out. Duplicate-aware grouping keeps no held-out song's twin in training.

Table 1. Corpus, deduplicated by stem and lyrics hash; the batches share one song, so the pool is 71 rather than 72, and filenames alone give 110 apparent songs, an inflation of 1.55× that any pipeline trusting filenames inherits. Batch 2 supplied the 114 clips used in §3.3; like the rest, they are model-generated.

	Batch 1	Batch 2	Pooled
Clips	61	114	175
Filename stems	20	91	110
Distinct songs (deduplicated)	19	53	**71**
Hours	3.66	7.29	10.95
Caption style	prose	tag-list	—

2.2 Adaptation and generation

We adapted the 2B `base` variant with a LoRA [16] (rank 128, α 256, targeting attention and MLP projections), warm-started from an earlier checkpoint trained on batch 1.

Evaluation used 60 fixed prompts in three families of 20: held-out lyrics with their own caption, with a templated genre caption, and with a templated caption over another song's lyrics. Seeds came from the prompt identifier by CRC-32 and were matched take-for-take, so prompt p take t used an identical seed under every setting. We generated $k = 3$ takes per prompt, raised to $k = 5$ for the primary contrast after pilot variance made the intraclass correlation (0.34) impossible to ignore. Conditions span a guidance sweep of $\{1, 2, 3, 5, 7, 10, 15\}$ plus baselines with and without the adapter on `base` and distilled `turbo`: 3,214 generations total.

2.3 Metrics

Lyric intelligibility. We separated vocals with Demucs [13], transcribed with MMS-1b-all [12] using its isiZulu adapter, and compared against the prompt lyrics. Character error rate suits isiZulu's conjunctive orthography better than word error rate.

Bare CER covers only attempted lines. Alignment is caused by the treatment, so conditioning on it conditions on a post-treatment collider. A *composite* error charges unattempted lines the maximum of 1.0,

$$\text{composite} = \text{coverage} \times \text{CER} + (1 - \text{coverage}) \times 1.0,$$

retaining all 60 prompts with no selection assumption. Coverage, the fraction of prompt lines attempted, is reported separately: a model that sings nothing scores well on CER alone.

Genre adherence. We ranked each generation's true caption against 39 sampled distractors using LAION-CLAP [11] (HTSAT-base, music-domain checkpoint). Nominal chance top-1 is 0.025, but ranks use a strict inequality, so duplicates tie for free. Our 60 prompts carry only 20 distinct captions: an

undeduplicated pool holds roughly five free copies of the target and effective chance is nearer 0.069. We deduplicate for the corpus re-embedding of §3.3; every generated-audio retrieval figure here predates that fix and should be read against the higher rate.

2.4 Statistical protocol

The unit of analysis is the prompt, not the generation. The intraclass correlation is 0.34 for genre and 0.27 for CER, so within-prompt seed variance exceeds between-prompt variance. Takes are therefore averaged within prompt, and tests pair over prompts.

Prompts are not independent either, which bounds every number we report. The 60 prompts draw on 20 distinct captions and 15 distinct lyric sets, so the effective sample is 20 for a caption-keyed measurement and 15 for a lyric-keyed one. We report every test twice, over prompts and clustered, and quote the clustered figure wherever the two disagree. The headline guidance result survives the stricter unit: composite lyric error falls by 0.406 clustered over the 15 lyric sets ($p = 4 \times 10^{-5}$), against 0.437 over prompts.

We abandoned a bootstrap CI-inversion statistic after finding it anticonservative roughly threefold ($p = 0.0011$ against a permutation reference's 0.0034). All p-values are two-sided sign-flip permutation tests over 200,000 permutations, which the matched-seed design justifies exactly. Intervals are percentile bootstrap over prompts. The analysis plan, prompts and splits were registered and hashed before any generation ran. One declared deviation: the primary endpoint was registered over two families ($n = 40$), while an initial analysis used all three.

2.5 Compute

Adaptation ran on a rented NVIDIA RTX 6000 Ada (48 GB) at \$0.78/hour: about 16.4 hours of wall-clock from checkpoint timestamps, over 400 epochs and 4,250 optimiser steps, roughly \$13. Evaluation ran on an RTX 4000 Ada (20 GB) at \$0.28/hour, averaging 11 seconds per 60-second clip, so 3,214 generations took about 9.8 GPU-hours, roughly \$3. Adaptation cost nearly five times the evaluation and bought no measurable lyric benefit. Diagnosing and repairing the retrieval control, the widest-reaching finding, cost about a dollar and no new generations. Nothing here needs a cluster.

3 Results

3.1 Guidance controls whether the model sings

Composite lyric error falls from 0.957 at guidance 1 to 0.459 at 15, with slope -0.203 per unit log-guidance (95% CI $[-0.222, -0.184]$). Coverage rises from 0.072 to 0.680, and ordering is perfect both ways (Spearman $\rho = \mp 1.000$), though seven level means are weak evidence alone (Figure 1(a)). On the tested contrast, guidance 1 against 7, composite error differs by -0.437 ($d_z = -2.57$, $p < 10^{-5}$).

The magnitude is the finding, not the direction. At guidance 1.0 the adapted model attempts 7% of prompt lyric lines. Listening to those takes ourselves, unblinded, phonotactics, vowels and melisma are right, but the words are not words. Tuning on audio quality alone, without intelligibility in the loop, will optimise the system into unintelligibility.

Our pre-registered primary endpoint was binarised top-1 caption retrieval, conditional on the instrument clearing a usability gate, which it passed at 0.25; it gives $+0.105$ ($p = 0.019$). The repaired instrument (§3.3) puts that gate at 0.044 with a gap interval spanning zero, so the endpoint is void. The broken instrument is itself a finding, not a defeat. The lyric axis, a registered secondary, is what this paper reports.

We registered the prediction that low guidance would damage style conditioning while sparing lyrics, with an equivalence margin of ± 0.05 CER declared in advance. It is refuted: the difference is -0.186, nearly four times outside the margin and opposite in direction.

Three controls bound the reading. The same ASR pipeline over the corpus's held-out audio gives CER 0.248 against 0.242 at guidance 7, so the model is no worse than the material it was adapted on. Disabling the vocal separator reproduces the sweep (0.242 against 0.400 at guidance 1), so the effect is not an artefact of a separator trained on Western pop. Vocal prominence, the obvious loudness confound, accounts for only 6% of the association (95% CI $[2\%, 10\%]$), estimated over takes rather than prompts and therefore treated as indicative.

3.2 The adaptation does not carry the effect

If guidance rather than adaptation governs lyric-following, the adapter should be removable without much cost. It is. Holding variant, steps, shift and guidance at 7.0 and removing only the LoRA, no lyric metric improves: composite $+0.027$, coverage -0.032, bare CER $+0.014$ ($p = 0.045$, 0.058, 0.033, adapted minus unadapted over 60 prompts). We do not read this as harm: these are three views of one comparison, none surviving Bonferroni or clustering at the 15 lyric sets (all $p \geq 0.10$). Adaptation bought no measurable improvement where guidance moved the same metric by 0.437. The contrast used one guidance value, so we cannot exclude an interaction at lower settings.

The null is architecturally predicted, not merely observed: LoRA injection walks the decoder only, so the encoder that turns isiZulu characters into the representation the decoder attends to is never adapted, at any rank. The adapter also lowers caption retrieval on both variants, a property of the measurement rather than the model: adapting towards isiZulu moves audio away from an embedding trained predominantly on Western material.

The distilled variant discards the control. `turbo` is distilled without classifier-free guidance [10]: it accepts a `guidance_scale` argument, overrides it to 1.0 before inference, and defaults to 8 steps. None of this is hidden: the technical report says the distilled model runs "without Classifier-Free Guidance". But it is not surfaced to the caller, and `turbo` is the default checkpoint. Across 900 clips at five nominal values the override fired on all 720 requesting another, logging

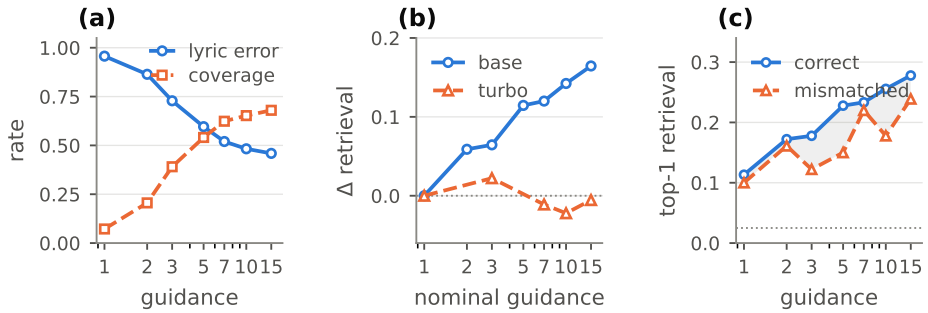


Fig. 1. (a) On `base`, composite lyric error falls and coverage rises monotonically with guidance; the tested contrast is -0.437, $d_z = -2.57$, $n = 60$. (b) Change in caption retrieval from each variant's own guidance-1 value, plotted as change because the variants run 50 and 8 steps: `base` responds, `turbo` is flat. (c) The contaminated control as originally scored; the arms track because 67–80% of the "mismatched" trials carried a byte-identical caption.

each time and surfacing on none, and retrieval does not order with the nominal value where `base` responds (Figure 1(b)). Sweeping guidance on a default install sweeps an inert parameter. What is lost is the control, not the intelligibility: `turbo` reaches coverage 0.507 and CER 0.263 at its overridden setting, close to `base` at guidance 7 and far from guidance 1.

3.3 How we broke the retrieval control, three times

Our caption-retrieval control reported that LAION-CLAP could not tell a correct isiZulu caption from a mismatched one. The report was wrong for three reasons, each invisible in its output.

First, the mismatched caption was taken as the next item in list order. Our manifests are prompt-major, so that neighbour is usually another take of the same prompt with a byte-identical caption. Thus, 67–80% of negative-control trials on generated audio were positive trials wearing a negative label.

Second, the clip's own caption remained in the distractor pool. Whenever the embedding worked, it ranked the correct caption above the target it was scored against. This pushed the negative arm *below* chance in proportion to the embedding's real ability.

Third, we scored binarised top-1 hits and discarded the ranks, which costs roughly fourfold in efficiency here. All three errors push the same way, towards a false negative, and none of them raises an error.

Corrected, the instrument discriminates. Re-embedding the 114 corpus clips with a deduplicated pool that bars every copy of the clip's own caption gives a correct-caption rank 6.57 places above a mismatched one (95% CI $[3.00, 10.08]$, $p = 0.0008$, over the 63 distinct captions; 5.75 over the 114 clips). Top-1 is 0.044 against chance 0.025 and does not separate. The signal is in the rank.

On generated audio, the clips are no longer available to re-embed. We therefore restrict the control to trials whose paired caption genuinely differs: the last take of each prompt, $n = 59$. The correct caption ranks above the mismatched one at six of seven guidance levels, with four surviving Bonferroni. The same six survive on RMS-normalised audio, so this is not the vocal merely becoming louder. The binarised statistic finds two (Figure 1(c)). This arm fixes only the first and third defects. Its rank gap is therefore amplified by the embedding's own competence and is evidence that discrimination exists, never of its size. The retained take also differs between $k = 3$ and $k = 5$, so the seven levels are not a clean dose-response.

The corpus captions are the prompts that generated the audio, so this is the most favourable case available, not an independent test. It is also not a specificity estimate. Once the clip's own caption is barred, the negative arm is no longer a chance baseline, which is why the Rogan–Gladen correction we attempted returned values above 1.0.

4 Impact in African Contexts

1. **Sweep the knob before you train the adapter.** Guidance, not the LoRA, controls whether the model sings the requested isiZulu words. Our $13 adaptation bought no measurable lyric benefit; the $3 sweep did.
2. **Count your training examples at the last stage of the pipeline.** In the v1 pipeline, `py3langid` classified isiZulu lyrics as isiXhosa (85% shared vocabulary) and sometimes as Swahili. Neither code was supported, so the loader raised `ValueError` and moved on: no error in the logs, a normal loss curve, and some songs never reached training. Any pipeline treating an unsupported code as a per-item exception has this bug.
3. **Count what you actually have.** Deduplicate by content, not filename: filename grouping overstated our corpus by 1.55×. Audit transcripts for cross-language hallucination. And count distinct captions and lyrics, not prompts: our 60 prompts drew on 20 captions and 15 lyric sets.
4. **Validate your metric, then check the control itself.** Take roughly 100 in-language clips with captions, measure retrieval against the correct and a mismatched caption, and compare both to chance; if the gap is under a point, the metric is not measuring what you think. Then check the control against the three defects of §3.3: ours failed all three. If validation audio is model-generated, do not validate on it, because clips generated *from* those captions cannot separate a weak metric from a weak generator.
5. **Consent does not disappear when the audio is synthetic.** A generator prompted for Isicathamiya or Maskandi imitates a tradition from an undisclosed corpus, moving the question upstream to a party you cannot ask. South African law vests certain rights in communities rather than individuals [18, 17]: no one artist can clear a tradition.

A pipeline that reports on the step rather than the artefact cannot tell you the difference.

5 Limitations

All training audio comes from one artist, so held-out clips share caption conventions with training clips. It was generated on a commercial service under a paid subscription whose terms assign the subscriber ownership of the output but prohibit training other models on it; our LoRA training falls under that prohibition. We redistribute no source audio; the subscriber is a coauthor and consented. The adapter's contribution over guidance is null here, so our positive finding does not depend on the adaptation having worked.

The lyric metric depends on MMS-1b-all, whose isiZulu accuracy is imperfect. Matched seeds and an identical pipeline mean ASR error should attenuate rather than manufacture the trend. Absolute values are not intelligibility rates.

The repaired retrieval metric validates the instrument but cannot rank systems ($d_z = 0.34$, 69 of 114 clips). The artist who supplied the corpus, a first-language isiZulu speaker, compared all 60 prompts blind, `base` at guidance 7 against `turbo`. She preferred `turbo` for intelligibility (17 to 4, sign test $p = 0.007$) and quality (25 to 11, $p = 0.029$), and split on authenticity (9 to 9). Our ASR metric orders the two the other way (CER 0.242 against 0.263); we report the disagreement rather than resolve it. But $n = 1$, from the person who supplied the data, is one reading, not a listening study: the main gap here.

A withdrawal mechanism was designed but never built, and revocation does not reach trained weights, so withdrawal is a governance fiction unless the mechanism precedes training. We make no claim about isiZulu music in general.

6 Conclusion

We adapted an open text-to-music model to isiZulu, and the three findings point one way. First, adaptation was not what made it sing: guidance was. Composite lyric error moves by 0.437 between guidance 1 and 7, the effect survives clustering at the 15 distinct lyric sets, and the adapter improves no lyric metric at matched guidance. Second, the checkpoint most people run discards that control. The distilled `turbo` variant overrides the guidance argument to 1.0, logs the override, and surfaces nothing to the caller, so a default install sweeps an inert parameter. Third, the caption-retrieval control we built to catch this was itself broken in three independent ways, each pushing towards a false negative we believed for weeks, because a plausible negative result about an under-resourced language is the one nobody interrogates.

Three failures ran silently across this work: a language classifier, a serving override, and our own analysis code, whose defects are the third finding above. Not one raised an error.

The rule this leaves is cheap. Sweep the inference control before spending on adaptation, and validate the metric before trusting either: our $13 adaptation bought no measurable lyric benefit, the $3 sweep did. What we most want replicated is the same sweep on field recordings, with more than one listener.

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